

ORIGINAL RESEARCH ARTICLE

Predicting the Market Microstructure–Price Efficiency Nexus Using Time Series Analysis in Nigeria

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Abstract

This study investigates the predictive relationship between market microstructure and price efficiency in the Nigerian stock market using time series analysis over the period 1995–2023. Market microstructure, operationalized through bid-ask spread, trading volume, and market depth, is examined as a determinant of price efficiency, measured by the variance ratio test statistic. The study employs an Autoregressive Distributed Lag (ARDL) bounds testing approach, controlling for gross domestic product per capita growth rate, inflation rate, population growth rate, trade openness, institutional quality proxied by the corruption perception index, natural resource dependence measured as oil revenue to total revenue ratio, human capital proxied by life expectancy, and infrastructure development captured through access to electricity. Results from the time series regression reveal that bid-ask spread exerts a statistically significant negative effect on price efficiency in both the short run and the long run, while trading volume and market depth positively predict improvements in price efficiency. Among the control variables, GDP per capita growth, trade openness, institutional quality, and infrastructure development demonstrate significant positive associations with price efficiency, whereas natural resource dependence and inflation rate exhibit negative effects. Post-estimation diagnostics confirm the stability, normality, and serial correlation-free nature of the estimated model. The findings contribute to the growing body of literature on market microstructure in emerging markets by providing granular, time-varying evidence from Nigeria—a frontier market characterized by structural peculiarities that distinguish it from developed economies. The study recommends targeted regulatory interventions to enhance market transparency, reduce transaction costs, and strengthen institutional frameworks as pathways to improving informational efficiency in the Nigerian capital market.

Keywords: Market microstructure, price efficiency, time series analysis, ARDL, Nigerian stock market, bid-ask spread, emerging markets

JEL Classification: G10, G14, G15, C22

1. Introduction

The efficient market hypothesis, first formalized by Fama (1970), posits that asset prices in financial markets fully reflect all available information, implying that no investor can consistently achieve abnormal returns through trading strategies based on publicly available data. This foundational proposition has shaped decades of financial research, regulatory design, and investment practice across global capital markets. However, the theoretical elegance of the efficient market hypothesis has been progressively challenged by empirical evidence documenting persistent anomalies, predictable patterns, and systematic deviations from informational efficiency, particularly in emerging and frontier markets (Lo, 2004; Lim & Brooks, 2011). The recognition that real-world markets deviate from the idealized frictionless conditions assumed by classical finance has given rise to the field of market microstructure—the study of the processes and outcomes of exchanging assets under explicit trading rules (O'Hara, 1995).

Market microstructure theory provides the analytical architecture for understanding how the mechanics of trading—order flow, bid-ask spreads, market depth, transaction costs, information asymmetry, and the behavior of market makers—interact to determine the speed, accuracy, and completeness with which prices incorporate new information (Madhavan, 2000). Unlike classical asset pricing models that treat markets as abstract aggregation mechanisms, microstructure theory recognizes that the institutional design of trading platforms, the strategic behavior of informed and uninformed traders, and the regulatory environment collectively shape the informational content of prices (Hasbrouck, 2007). In this sense, market microstructure operates as the transmission mechanism through which information flows into prices, making it a critical determinant of price efficiency.

The Nigerian stock market, operated through the Nigerian Exchange Group (formerly the Nigerian Stock Exchange), represents a compelling empirical context for investigating the microstructure–price efficiency nexus. As Africa's largest economy by GDP and one of the continent's most capitalized equity markets, Nigeria occupies a strategic position in the global financial architecture. Yet the Nigerian stock market exhibits structural characteristics—low liquidity, high concentration, limited market-making capacity, information asymmetry, and regulatory fragility—that distinguish it fundamentally from the developed markets where the bulk of microstructure research has been conducted (Adelegan, 2009; Okpara, 2010). These structural features create a natural laboratory for examining how microstructural frictions impede or facilitate the price discovery process, and whether improvements in market design can enhance informational efficiency over time.

The motivation for this study stems from several converging considerations. First, the empirical literature on market microstructure remains overwhelmingly concentrated in developed market settings, particularly the United States and Europe, where high-frequency data and sophisticated trading infrastructure enable granular analysis (Biais, Glosten, & Spatt, 2005). The applicability of insights derived from these contexts to emerging and frontier markets characterized by fundamentally different institutional environments remains an open empirical question. Second, Nigeria has undergone significant capital market reforms since the early 2000s, including the adoption of automated trading systems, the dematerialization of securities, and the introduction of market-making arrangements. Understanding whether and how these reforms have translated into measurable improvements in price efficiency requires rigorous time series

analysis that accounts for structural breaks and evolving microstructural conditions. Third, the macroeconomic environment in which the Nigerian stock market operates is characterized by volatility in oil revenues, inflationary pressures, exchange rate fluctuations, and institutional challenges that interact with microstructural variables in complex ways. Isolating the microstructure effect on price efficiency, therefore, requires careful specification of control variables that capture these macroeconomic and institutional dynamics.

Despite the theoretical centrality of market microstructure to price efficiency, the empirical evidence linking specific microstructural variables to measurable dimensions of price efficiency in the Nigerian context remains sparse, fragmented, and methodologically inconsistent. Existing studies have predominantly adopted cross-sectional or event-study approaches that capture snapshot relationships rather than the dynamic, time-varying nature of the microstructure–efficiency nexus (Ajao & Osayuwu, 2012; Nwidobie, 2014). The few time series studies that exist have typically employed rudimentary econometric techniques—ordinary least squares regression or simple unit root–based efficiency tests—that fail to account for the cointegrating relationships, structural breaks, and short-run dynamics that characterize macroeconomic and financial time series in Nigeria.

Furthermore, the existing literature has inadequately addressed the role of macroeconomic and institutional control variables in conditioning the microstructure–efficiency relationship. Market microstructure does not operate in a vacuum; it is embedded within broader economic, institutional, and infrastructural contexts that shape market participation, information production, and investor behavior. Omitting relevant control variables introduces specification bias and renders empirical estimates unreliable for policy inference. The specific research problem this study addresses is therefore: What is the predictive relationship between market microstructure variables and price efficiency in the Nigerian stock market over time, after controlling for macroeconomic, institutional, and structural factors?

This study makes several distinct contributions to the scholarly literature and to policy discourse. First, it provides the most comprehensive time series analysis of the market microstructure–price efficiency nexus in Nigeria to date, spanning nearly three decades (1995–2023) and incorporating multiple dimensions of microstructure that are rarely examined simultaneously. Second, the study introduces a robust set of control variables—GDP per capita growth rate, inflation rate, population growth rate, trade openness, institutional quality, natural resource dependence, human capital, and infrastructure development—that capture the multidimensional environment within which the Nigerian stock market operates. This comprehensive specification enhances the internal validity of the estimates and provides a more nuanced understanding of the factors that jointly determine price efficiency. Third, the adoption of the ARDL bounds testing approach represents a methodological advancement over previous studies, as it permits the estimation of both short-run and long-run relationships without requiring that all variables be integrated of the same order, thereby accommodating the mixed integration properties typically observed in Nigerian economic data. Fourth, the study generates actionable policy recommendations grounded in empirical evidence, contributing to ongoing capital market development efforts in Nigeria and, by extension, in comparable emerging market economies.

This study is delimited to the Nigerian stock market over the period 1995 to 2023, corresponding to the period for which consistent data on all variables of interest are

available. The analysis focuses on the aggregate market level rather than individual securities, consistent with the macrostructural perspective adopted. Price efficiency is measured using the variance ratio test statistic, which captures the degree to which stock returns approximate a random walk. Market microstructure is operationalized through three aggregate proxies: the average bid-ask spread, total market trading volume, and market depth. The study does not examine intraday or high-frequency microstructural dynamics, as the data resolution required for such analysis is not consistently available for the Nigerian market over the full sample period.

2. Literature Review and Hypothesis Development

2.1 Conceptual Literature: Definitions

Market Microstructure refers to the branch of financial economics that studies the process by which investors' latent demands are ultimately translated into prices and traded volumes. O'Hara (1995, p. 1) defined it as "the study of the process and outcomes of exchanging assets under explicit trading rules." The concept encompasses the mechanisms of trade execution, the formation of bid and ask prices, the role of intermediaries such as market makers and specialists, the dynamics of order flow, and the information content embedded in trading activity. Key microstructural variables include the bid-ask spread (the difference between the price at which a dealer is willing to sell and willing to buy a security), trading volume (the total quantity of shares exchanged over a given period), and market depth (the ability of the market to absorb large orders without significant price impact).

Price Efficiency refers to the degree to which market prices reflect the true intrinsic value of assets based on all available information. Following Fama (1970), three forms of efficiency are traditionally distinguished: weak-form efficiency (prices reflect all past trading information), semi-strong efficiency (prices reflect all publicly available information), and strong-form efficiency (prices reflect all information, including private information). In empirical research, price efficiency is most commonly tested through the random walk hypothesis, which posits that price changes are serially uncorrelated and unpredictable. The variance ratio test, introduced by Lo and MacKinlay (1988), provides a statistical measure of the degree to which a price series conforms to the random walk, with values closer to unity indicating greater efficiency.

2.2 Theoretical Literature Review

Several theoretical frameworks underpin the relationship between market microstructure and price efficiency.

The Information Asymmetry Model (Kyle, 1985; Glosten & Milgrom, 1985): Kyle's (1985) sequential trade model demonstrates that informed traders strategically manage their order flow to exploit private information, while market makers adjust prices to reflect the expected information content of incoming orders. The bid-ask spread emerges as a compensation mechanism for market makers who face adverse selection risk from trading with better-informed counterparties. Glosten and Milgrom (1985) formalized this adverse selection component of the spread, showing that wider spreads reflect greater information asymmetry and, paradoxically, can impede the speed of price adjustment by deterring uninformed liquidity providers. Together, these models establish a direct

theoretical link between microstructural variables (particularly the bid-ask spread) and the efficiency of the price discovery process.

The Inventory Model (Stoll, 1978; Ho & Stoll, 1981): The inventory-based approach to market microstructure emphasizes the role of market makers' inventory management in determining bid-ask spreads and price dynamics. Market makers who accumulate unwanted inventory positions adjust their quotes to attract offsetting order flow, creating temporary price deviations from fundamental values. In markets with thin trading and concentrated ownership—features characteristic of the Nigerian stock market—inventory effects can be particularly pronounced, creating persistent pricing inefficiencies.

The Noise Trader Theory (Black, 1986; De Long, Shleifer, Summers, & Waldmann, 1990): Black (1986) distinguished between information-motivated and noise-motivated trading, arguing that noise traders create liquidity that enables markets to function but simultaneously introduce pricing errors. De Long et al. (1990) formalized this insight, demonstrating that noise trader risk can cause prolonged deviations of prices from fundamental values, particularly when arbitrage is costly or risky. In markets with a high proportion of retail, uninformed investors—as is the case in Nigeria—noise trading may dominate the price formation process, reducing informational efficiency.

The Adaptive Markets Hypothesis (Lo, 2004): Lo's adaptive markets hypothesis provides an evolutionary framework that reconciles the efficient market hypothesis with behavioral finance. According to this perspective, market efficiency is not a fixed property but varies over time as market participants adapt to changing environments, new technologies, and regulatory reforms. This theoretical lens is particularly relevant for the Nigerian context, where ongoing structural reforms are expected to produce time-varying improvements in market efficiency.

2.3 Conceptual Framework

The conceptual framework of this study posits market microstructure as the primary independent variable influencing price efficiency, with macroeconomic and institutional factors serving as control variables that condition the microstructure–efficiency relationship.

The framework operates through the following logic: Market microstructure variables—bid-ask spread, trading volume, and market depth—capture the transactional environment within which price discovery occurs. The bid-ask spread reflects information asymmetry and transaction costs; wider spreads signal greater frictions that impede the incorporation of information into prices, thereby reducing efficiency. Trading volume captures the intensity of information-motivated trading; higher volumes indicate greater participation and information processing, facilitating price discovery. Market depth captures the market's capacity to absorb orders without excessive price impact, supporting smooth price adjustment to new information.

These microstructural mechanisms operate within a broader macroeconomic and institutional context. GDP per capita growth reflects the overall economic dynamism that influences corporate earnings expectations and investor participation. Inflation erodes real returns and introduces noise into price signals. Trade openness captures the degree of international integration, which facilitates information flows and competitive pressures. Institutional quality shapes the regulatory environment, investor protection, and corporate governance standards that underpin market integrity. Natural resource dependence captures the structural vulnerability of oil-dependent economies to

commodity price shocks. Human capital and infrastructure development influence the capacity of market participants to access, process, and act upon information.

2.4 Empirical Literature Review in Nigeria

The empirical literature on market efficiency and microstructure in Nigeria has evolved substantially over the past two decades, though significant gaps remain. Okpara (2010) tested the weak-form efficiency of the Nigerian stock market using serial correlation tests and runs tests on monthly All-Share Index returns from 1984 to 2009, finding evidence of predictable return patterns inconsistent with weak-form efficiency. Similarly, Ajao and Osayuwu (2012) employed the variance ratio test and found that Nigerian stock prices did not follow a random walk during the period 2001–2010, concluding that the market was informationally inefficient.

Nwidiobie (2014) examined the bid-ask spread as a measure of transaction costs in the Nigerian stock market, documenting persistently wide spreads attributable to low liquidity, concentrated ownership, and limited market-making activity. The study concluded that high transaction costs represented a significant barrier to price efficiency. Ogege and Ezike (2012) investigated the relationship between trading volume and stock price volatility, finding a significant positive relationship and suggesting that trading activity carried substantial information content in the Nigerian context.

More recently, Osamwonyi and Anikamadu (2020) examined the impact of capital market reforms on the efficiency of the Nigerian stock market, employing structural break tests alongside traditional efficiency metrics. Their findings suggested that the adoption of automated trading systems in 2013 was associated with measurable improvements in informational efficiency, though the market remained inefficient in absolute terms. Emenike and Okwuchukwu (2014) investigated the time-varying nature of stock market efficiency in Nigeria using rolling-window variance ratio tests, documenting cyclical patterns in efficiency that corresponded to macroeconomic and regulatory developments. Regarding the macroeconomic determinants of stock market development and efficiency, Yartey (2010) conducted a panel analysis of African stock markets, including Nigeria, and found that institutional quality, income level, and banking sector development were significant predictors of market capitalization and liquidity. Oluwatobi and Ogunrinola (2011) examined the relationship between human capital development and economic growth in Nigeria, providing indirect evidence on the channels through which education and health investments influence financial market participation and efficiency.

2.5 Critical Analysis of the Literature

A critical examination of the existing literature reveals several limitations that constrain the current understanding of the microstructure–price efficiency relationship in Nigeria. First, most studies have tested market efficiency in isolation, without explicitly linking efficiency outcomes to specific microstructural determinants. The typical research design tests whether prices follow a random walk but does not investigate why deviations from efficiency occur or which microstructural features are most consequential. This disconnect between efficiency testing and microstructure analysis limits the policy relevance of the findings.

Second, the econometric approaches employed in many existing studies are insufficiently sophisticated for the task at hand. The use of standard OLS regression for time series data

without adequate attention to stationarity, cointegration, and dynamic specification produces potentially spurious results. Few studies employ cointegration techniques or error correction models that would permit the identification of both short-run and long-run relationships.

Third, the treatment of control variables in existing studies is inadequate. Most analyses of Nigerian stock market efficiency either omit macroeconomic controls entirely or include them in a perfunctory manner without theoretical justification. The omission of institutional quality, natural resource dependence, infrastructure development, and human capital variables is particularly problematic given the structural characteristics of the Nigerian economy.

Fourth, the literature suffers from a temporal limitation. Many studies rely on data samples that end before the significant reforms and structural changes that occurred in the Nigerian capital market after 2010, including the introduction of automated trading, the adoption of international financial reporting standards, and the reorganization of the Nigerian Stock Exchange into the Nigerian Exchange Group.

2.6 Research Gaps

Based on the critical analysis above, this study identifies the following research gaps:

- i. There is a paucity of studies that explicitly model the predictive relationship between market microstructure variables and price efficiency in Nigeria using appropriate time series techniques.
- ii. Existing studies do not adequately control for the macroeconomic, institutional, and structural variables that condition the microstructure–efficiency relationship.
- iii. The dynamic, time-varying nature of the microstructure–efficiency nexus has not been captured through cointegration and error correction frameworks.
- iv. The literature lacks evidence from the most recent period (post-2015) during which significant regulatory and structural changes have occurred in the Nigerian capital market.

2.7 Hypotheses

Based on the theoretical and empirical review, the following hypotheses are formulated:

H₁: Bid-ask spread has a significant negative predictive effect on price efficiency in the Nigerian stock market.

H₂: Trading volume has a significant positive predictive effect on price efficiency in the Nigerian stock market.

H₃: Market depth has a significant positive predictive effect on price efficiency in the Nigerian stock market.

3. Methodology

3.1 Research Design

This study adopts a quantitative, longitudinal research design employing annual time series data spanning 1995 to 2023, yielding 29 observations. The time series approach is appropriate for capturing the dynamic, evolving relationship between market microstructure and price efficiency over time. The choice of annual frequency is dictated by the availability of consistent data on the control variables, several of which (GDP per

capita growth, institutional quality, infrastructure development) are reported only at annual intervals. While higher-frequency data would enable more granular microstructural analysis, the annual frequency permits the integration of macroeconomic and institutional variables into a comprehensive model specification that would not be feasible at daily or monthly frequency.

3.2 Model Specification

The study employs the Autoregressive Distributed Lag (ARDL) bounds testing approach developed by Pesaran, Shin, and Smith (2001). This approach offers several advantages for the present study. First, it accommodates variables with different orders of integration— $I(0)$, $I(1)$, or a mixture thereof—without requiring pre-testing for unit roots (though unit root tests are conducted for robustness). Second, it simultaneously estimates short-run dynamics and the long-run equilibrium relationship. Third, it performs well in small sample sizes, which is relevant given the 29-year sample period.

The baseline ARDL model is specified as follows:

Long-run equation:

$$PE_t = \alpha_0 + \beta_1 BAS_t + \beta_2 TV_t + \beta_3 MD_t + \gamma_1 GDPPC_t + \gamma_2 INF_t + \gamma_3 POP_t + \gamma_4 TRADE_t + \gamma_5 INST_t + \gamma_6 NRD_t + \gamma_7 HC_t + \gamma_8 INFRA_t + \varepsilon_t$$

Whereas:

- PE_t = Price Efficiency (variance ratio test statistic) at time t
- BAS_t = Bid-Ask Spread (average annual)
- TV_t = Trading Volume (log of total annual trading volume)
- MD_t = Market Depth (ratio of market capitalization to GDP)
- $GDPPC_t$ = GDP per capita growth rate
- INF_t = Inflation rate (consumer price index, annual percentage change)
- POP_t = Population growth rate
- $TRADE_t$ = Trade openness (exports + imports as a percentage of GDP)
- $INST_t$ = Institutional quality (Corruption Perception Index score)
- NRD_t = Natural resource dependence (oil revenue as a percentage of total government revenue)
- HC_t = Human capital (life expectancy at birth)
- $INFRA_t$ = Infrastructure development (percentage of population with access to electricity)
- ε_t = Error term

The ARDL error correction model (ECM):

$$\Delta PE_t = \alpha_0 + \sum_{i=1}^p \varphi_i \Delta PE_{t-i} + \sum_{j=0}^q \delta_{1j} \Delta BAS_{t-j} + \sum_{j=0}^q \delta_{2j} \Delta TV_{t-j} + \sum_{j=0}^q \delta_{3j} \Delta MD_{t-j} + \sum_{j=0}^q \psi_{1j} \Delta GDPPC_{t-j} + \sum_{j=0}^q \psi_{2j} \Delta INF_{t-j} + \sum_{j=0}^q \psi_{3j} \Delta POP_{t-j} + \sum_{j=0}^q \psi_{4j} \Delta TRADE_{t-j} + \sum_{j=0}^q \psi_{5j} \Delta INST_{t-j} + \sum_{j=0}^q \psi_{6j} \Delta NRD_{t-j} + \sum_{j=0}^q \psi_{7j} \Delta HC_{t-j} + \sum_{j=0}^q \psi_{8j} \Delta INFRA_{t-j} + \lambda ECM_{t-1} + \varepsilon_t$$

Where λ represents the speed of adjustment coefficient, which should be negative and significant for the error correction mechanism to be valid.

3.3 Variables and Their Measures

Table 1: Variables, Definitions, and Measurement

Variable	Symbol	Definition	Measurement
Price Efficiency	PE	Degree to which stock prices follow a random walk	Variance ratio test statistic (value closer to 1 = higher efficiency)
Bid-Ask Spread	BAS	Average difference between ask and bid prices	Annual average proportional bid-ask spread (%)
Trading Volume	TV	Total quantity of shares traded	Natural logarithm of annual trading volume (million shares)
Market Depth	MD	Market's capacity to absorb large orders	Market capitalization as % of GDP
GDP per Capita Growth	GDPPC	Growth in per capita economic output	Annual % change in real GDP per capita
Inflation Rate	INF	Rate of price level increase	Annual % change in Consumer Price Index
Population Growth Rate	POP	Rate of population increase	Annual % change in total population
Trade Openness	TRADE	Degree of international trade integration	$(\text{Exports} + \text{Imports}) / \text{GDP} \times 100$
Institutional Quality	INST	Quality of governance and anti-corruption	Corruption Perception Index score (0–100)
Natural Resource Dependence	NRD	Reliance on oil revenue	$\text{Oil revenue} / \text{Total government revenue} \times 100$
Human Capital	HC	Quality of human resources	Life expectancy at birth (years)
Infrastructure Development	INFRA	Physical infrastructure availability	% of population with access to electricity

3.4 A Priori Expectations

Table 2: A Priori Expectations

Variable	Expected Sign	Theoretical Justification
BAS	Negative (–)	Higher spreads indicate greater transaction costs and information asymmetry, impeding price discovery (Glosten & Milgrom, 1985)
TV	Positive (+)	Higher trading volume facilitates information incorporation into prices (Kyle, 1985)
MD	Positive (+)	Deeper markets absorb information shocks more smoothly, enhancing efficiency (O'Hara, 1995)
GDPPC	Positive (+)	Economic growth enhances corporate fundamentals and investor participation (Yartey, 2010)
INF	Negative (–)	Inflation introduces noise into price signals and reduces real returns (Fama & Schwert, 1977)
POP	Ambiguous (±)	Higher population growth may increase market participation (+) or strain resources (–)
TRADE	Positive (+)	Trade openness facilitates information flows and competitive pressures (Bekaert et al., 2005)
INST	Positive (+)	Better institutions enhance investor protection and market integrity (La Porta et al., 1998)
NRD	Negative (–)	Oil dependence introduces volatility and diverts economic focus from financial development (Sachs & Warner, 2001)
HC	Positive (+)	Better human capital improves information processing and financial literacy (Romer, 1990)
INFRA	Positive (+)	Better infrastructure facilitates market access and information dissemination (Roller & Waverman, 2001)

3.5 Data Sources

Data for this study are sourced from multiple authoritative databases. Stock market variables (bid-ask spread, trading volume, market capitalization) are obtained from the Nigerian Exchange Group annual reports and the Securities and Exchange Commission (SEC) Nigeria statistical bulletins. The All-Share Index data for constructing the variance ratio test statistic are sourced from the Central Bank of Nigeria (CBN) Statistical Bulletin. GDP per capita growth, inflation rate, population growth rate, trade openness, life expectancy, and access to electricity data are obtained from the World Bank's World Development Indicators (WDI) database. The Corruption Perception Index is sourced from Transparency International. Oil revenue data are obtained from the CBN Annual Report and Statement of Accounts.

3.6 Data Analysis Techniques

The empirical analysis follows a structured sequence:

- i. **Descriptive Statistics:** Computation of mean, median, standard deviation, minimum, maximum, skewness, and kurtosis for all variables to characterize the data distribution.
- ii. **Unit Root Tests:** Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests are conducted to determine the order of integration of each variable, ensuring that no variable is $I(2)$, which would invalidate the ARDL approach.
- iii. **ARDL Bounds Test for Cointegration:** The Pesaran et al. (2001) bounds test is employed to determine whether a long-run relationship exists among the variables. The computed F-statistic is compared against the critical value bounds.
- iv. **ARDL Long-Run and Short-Run Estimation:** Conditional on evidence of cointegration, the long-run coefficients and the error correction model are estimated. Optimal lag selection is determined using the Akaike Information Criterion (AIC).
- v. **Post-Estimation Diagnostics:** Serial correlation (Breusch-Godfrey LM test), heteroscedasticity (Breusch-Pagan-Godfrey test), normality (Jarque-Bera test), functional form (Ramsey RESET test), and stability (CUSUM and CUSUMSQ tests) are conducted.

3.7 Post-Estimation Checks

To ensure the robustness and reliability of the estimated model, the following diagnostic tests are performed: **Breusch-Godfrey Serial Correlation LM Test:** Tests for autocorrelation in the residuals, with the null hypothesis of no serial correlation. **Breusch-Pagan-Godfrey Heteroscedasticity Test:** Tests for non-constant variance of residuals, with the null hypothesis of homoscedasticity. **Jarque-Bera Normality Test:** Tests whether residuals are normally distributed, with the null hypothesis of normality. **Ramsey RESET Test:** Tests for functional form misspecification, with the null hypothesis of correct specification. **CUSUM and CUSUMSQ Tests:** Tests for structural stability of the estimated parameters over the sample period.

3.8 Significance Level

All hypothesis tests are evaluated at the 5% significance level ($\alpha = 0.05$), with robustness checks at the 1% and 10% levels also reported.

4. Results

4.1 Descriptive Statistics

Table 3: Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Min	Max	Skewness	Kurtosis	Obs.
PE	0.742	0.731	0.158	0.418	1.052	0.213	2.341	29
BAS	3.621	3.480	1.247	1.520	6.810	0.584	2.718	29
TV (log)	10.382	10.450	1.893	6.210	13.420	-0.321	2.156	29
MD	13.547	12.830	6.274	4.210	30.820	0.742	3.012	29
GDPPC	1.842	1.650	3.718	-4.230	10.470	0.418	3.241	29
INF	13.284	12.220	5.621	5.380	29.270	1.124	3.875	29
POP	2.584	2.590	0.087	2.410	2.720	-0.156	2.013	29
TRADE	34.217	33.450	9.834	15.670	53.280	0.098	2.312	29
INST	22.483	22.000	4.193	14.000	31.000	0.312	2.547	29
NRD	68.341	71.250	12.874	38.420	83.670	-0.687	2.618	29
HC	51.428	51.700	2.341	46.100	55.400	-0.452	2.234	29
INFRA	48.752	49.100	9.563	28.300	62.500	-0.278	2.089	29

Source: STATA Output

The descriptive statistics reveal several noteworthy patterns. The mean price efficiency value of 0.742 indicates that the Nigerian stock market, on average, deviates substantially from the random walk benchmark of 1.0, confirming a persistent degree of informational inefficiency over the sample period. The standard deviation of 0.158 suggests meaningful variation in efficiency across years, consistent with the adaptive markets perspective. The average bid-ask spread of 3.621% is remarkably high by international standards—the average spread on the New York Stock Exchange has typically been below 0.5% since the early 2000s—reflecting the substantial transaction cost barriers in the Nigerian market. Trading volume exhibits considerable variation (standard deviation of 1.893 in log terms), reflecting the episodic surges in trading activity associated with bull markets (2004–2008) and the subsequent crashes. Natural resource dependence averages 68.34%, confirming the dominance of oil revenues in Nigerian government finances. Institutional quality, as measured by the Corruption Perception Index, averages only 22.48 out of 100, placing Nigeria among the most challenging institutional environments globally.

4.2 Correlation Matrix

Table 4: Pairwise Correlation Matrix

	PE	BAS	TV	MD	GDPPC	INF	POP	TRADE	INST	NRD	HC	INFRA
PE	1.000											
BAS	-0.674	1.000										
TV	0.581	-0.432	1.000									
MD	0.523	-0.387	0.612	1.000								
GDPPC	0.384	-0.218	0.291	0.276	1.000							
INF	-0.412	0.367	-0.284	-0.312	-0.241	1.000						
POP	-0.187	0.142	-0.098	-0.074	-0.123	0.167	1.000					
TRADE	0.347	-0.267	0.312	0.289	0.198	-0.178	-0.087	1.000				
INST	0.478	-0.412	0.356	0.398	0.267	-0.312	-0.134	0.287	1.000			
NRD	-0.389	0.298	-0.241	-0.187	-0.156	0.214	0.098	0.178	-0.267	1.000		
HC	0.412	-0.321	0.387	0.356	0.178	-0.214	-0.312	0.198	0.434	-0.278	1.000	
INFRA	0.467	-0.378	0.421	0.389	0.213	-0.198	-0.267	0.234	0.512	-0.312	0.687	1.000

The correlation matrix reveals a strong negative correlation between bid-ask spread and price efficiency ($r = -0.674$), consistent with theoretical expectations. Trading volume ($r = 0.581$) and market depth ($r = 0.523$) exhibit moderate positive correlations with price efficiency. Among the control variables, institutional quality ($r = 0.478$), infrastructure development ($r = 0.467$), and human capital ($r = 0.412$) show the strongest positive associations with price efficiency. Natural resource dependence ($r = -0.389$) and inflation ($r = -0.412$) are negatively correlated with efficiency. The correlation between human capital and infrastructure ($r = 0.687$) suggests potential multicollinearity, which is monitored in the regression analysis through variance inflation factors.

4.3 Unit Root Tests

Before ARDL estimation, ADF and PP unit root tests were conducted. Results indicate that PE, GDPPC, and INF are stationary at levels— $I(0)$ —while BAS, TV, MD, TRADE, INST, NRD, HC, INFRA, and POP are stationary at first difference— $I(1)$. No variable is integrated of order two, confirming the appropriateness of the ARDL approach. The bounds test F-statistic of 5.847 exceeds the upper bound critical value of 4.68 at the 5% significance level (Pesaran et al., 2001, Table CI(iii), Case III), confirming the existence of a long-run cointegrating relationship among the variables.

4.4 Time Series Regression Results

Table 5: ARDL Long-Run and Short-Run Estimation Results

Panel A: Long-Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Decision
Constant	0.287	0.412	0.697	0.498	Insignificant
BAS	-0.048	0.012	-4.000	0.001	Significant***
TV	0.037	0.011	3.364	0.005	Significant***
MD	0.006	0.003	2.000	0.064	Significant*
GDPPC	0.009	0.004	2.250	0.041	Significant**
INF	-0.007	0.003	-2.333	0.034	Significant**
POP	-0.142	0.187	-0.759	0.461	Insignificant
TRADE	0.004	0.002	2.000	0.064	Significant*
INST	0.012	0.004	3.000	0.009	Significant***
NRD	-0.003	0.001	-3.000	0.009	Significant***
HC	0.008	0.007	1.143	0.272	Insignificant
INFRA	0.005	0.002	2.500	0.025	Significant**

*Note: ***, *, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Panel B: Short-Run (Error Correction) Coefficients

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Decision
Δ BAS	-0.032	0.009	-3.556	0.003	Significant***
Δ TV	0.024	0.008	3.000	0.009	Significant***
Δ MD	0.004	0.002	2.000	0.064	Significant*
Δ GDPPC	0.006	0.003	2.000	0.064	Significant*
Δ INF	-0.005	0.002	-2.500	0.025	Significant**
Δ POP	-0.087	0.134	-0.649	0.527	Insignificant
Δ TRADE	0.003	0.001	3.000	0.009	Significant***
Δ INST	0.008	0.003	2.667	0.018	Significant**
Δ NRD	-0.002	0.001	-2.000	0.064	Significant*
Δ HC	0.005	0.005	1.000	0.334	Insignificant
Δ INFRA	0.003	0.002	1.500	0.155	Insignificant
ECM(-1)	-0.683	0.124	-5.508	0.000	Significant***

$R^2 = 0.847$; Adjusted $R^2 = 0.789$; F-statistic = 14.628 ($p = 0.000$); Durbin-Watson = 2.087

The error correction term (ECM = -0.683) is negative, less than unity in absolute value, and highly significant, confirming a stable long-run equilibrium relationship. The coefficient indicates that approximately 68.3% of any disequilibrium in price efficiency is corrected within one year, suggesting a relatively rapid speed of adjustment. The model explains approximately 84.7% of the variation in price efficiency, and the overall F-statistic confirms the joint significance of the regressors.

4.5 Post-Estimation Tests

Table 6: Post-Estimation Diagnostic Tests

Test	Statistic	p-Value	Decision
Breusch-Godfrey Serial Correlation	$F(2,12) = 1.234$	0.324	No serial correlation
LM Test			
Breusch-Pagan-Godfrey	$F(11,14) = 0.876$	0.587	Homoscedastic residuals
Heteroscedasticity Test			
Jarque-Bera Normality Test	$JB = 1.842$	0.398	Normally distributed residuals
Ramsey RESET Test	$F(1,13) = 0.654$	0.433	No misspecification
CUSUM Test	Within 5% bounds	—	Stable parameters
CUSUMSQ Test	Within 5% bounds	—	Stable parameters

All diagnostic tests support the adequacy and reliability of the estimated model. The Breusch-Godfrey test fails to reject the null hypothesis of no serial correlation, the Breusch-Pagan-Godfrey test confirms homoscedasticity, and the Jarque-Bera test confirms normality of the residuals. The Ramsey RESET test provides no evidence of functional form misspecification. The CUSUM and CUSUMSQ plots remain within the 5% significance

bounds throughout the sample period, confirming the structural stability of the estimated parameters.

5. Discussion

5.1 Restatement of Key Findings

The empirical results provide robust evidence supporting the predictive relationship between market microstructure variables and price efficiency in the Nigerian stock market. The three core microstructure variables—bid-ask spread, trading volume, and market depth—all demonstrate statistically significant effects on price efficiency in the theoretically expected directions. The bid-ask spread emerges as the most powerful predictor of price efficiency, with a one-percentage-point increase in the average spread associated with a 0.048-unit decrease in the variance ratio statistic in the long run, representing a substantial deterioration in informational efficiency. Trading volume and market depth exert positive effects on efficiency, though the magnitude of the market depth coefficient is smaller and marginally significant in the long run.

Among the control variables, institutional quality and natural resource dependence demonstrate the most robust effects. A one-unit improvement in the Corruption Perception Index score is associated with a 0.012-unit improvement in price efficiency, while a one-percentage-point increase in the oil revenue share of government revenue reduces price efficiency by 0.003 units. Infrastructure development, GDP per capita growth, trade openness, and inflation also exhibit significant effects consistent with theoretical expectations. Population growth rate and human capital (life expectancy) do not achieve statistical significance, suggesting that their influence on price efficiency operates through indirect channels not fully captured by the model specification.

The error correction term of -0.683 indicates that the Nigerian stock market adjusts relatively quickly to disequilibria in price efficiency, with approximately two-thirds of any deviation from the long-run equilibrium corrected within a single year. This finding is noteworthy for a frontier market and suggests that, despite persistent structural inefficiencies, the market possesses meaningful self-correcting mechanisms.

5.2 Comparison with Previous Studies

The finding that bid-ask spreads negatively predict price efficiency is consistent with the theoretical models of Glosten and Milgrom (1985) and the empirical evidence from developed markets documented by Chordia, Roll, and Subrahmanyam (2008), who found that lower transaction costs were associated with greater market efficiency in the U.S. equity market. The magnitude of the effect documented in this study, however, is considerably larger than those typically reported for developed markets, reflecting the substantially wider spreads and more severe information asymmetry characteristic of the Nigerian market, as documented by Nwidobie (2014).

The positive effect of trading volume on price efficiency aligns with Kyle's (1985) theoretical prediction and with the empirical findings of Chordia, Subrahmanyam, and Anshuman (2001) for the U.S. market and Lesmond (2005) for a cross-section of emerging markets. The result also corroborates Ogege and Ezike's (2012) finding that trading volume carries significant information content in the Nigerian market.

The significance of institutional quality as a determinant of price efficiency resonates with the growing literature on the institutional foundations of financial market development.

La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1998) established the foundational link between legal institutions and financial outcomes, and more recently, Morck, Yeung, and Yu (2000) demonstrated that stock prices in countries with weaker institutions tend to move in lockstep, reflecting poor incorporation of firm-specific information—a hallmark of price inefficiency. The present study extends this literature by documenting the within-country, time-varying relationship between institutional quality and price efficiency in Nigeria.

The negative effect of natural resource dependence on price efficiency represents a novel contribution that connects the resource curse literature (Sachs & Warner, 2001) to financial market outcomes. The mechanism likely operates through multiple channels: oil revenue volatility introduces macroeconomic uncertainty that obscures fundamental signals; the concentration of economic activity in the extractive sector limits the diversity and informational richness of listed securities; and the political economy dynamics associated with resource dependence may undermine the regulatory independence necessary for effective capital market oversight.

5.3 Implications

The findings carry significant implications for both theory and practice. From a theoretical perspective, the results provide strong empirical support for the market microstructure approach to understanding price efficiency, validating the transmission mechanism posited by information asymmetry models in a frontier market context. The significance of the error correction term confirms that the microstructure–efficiency relationship is not merely a static association but a dynamic equilibrium that the market system tends to restore over time, consistent with the adaptive markets hypothesis.

From a practical and policy perspective, the findings suggest several priority areas for capital market regulators and policymakers. The dominant role of the bid-ask spread in predicting price efficiency points to the critical importance of policies that enhance market liquidity and reduce transaction costs. This includes strengthening market-making arrangements, reducing regulatory and operational barriers to trading, and promoting competitive fee structures among brokers and trading platforms. The significance of trading volume suggests that policies aimed at broadening market participation—financial literacy programs, simplified account opening procedures, technological innovation in trading platforms—can contribute to improvements in price efficiency. The importance of institutional quality underscores the complementarity between broader governance reforms and capital market development objectives.

5.4 Limitations

Several limitations should be acknowledged. First, the use of annual data, while necessary for incorporating macroeconomic control variables, inevitably smooths out the high-frequency dynamics of market microstructure that are central to the theoretical framework. Second, the aggregate market-level analysis may mask heterogeneity across individual securities, sectors, or market segments. Third, the measurement of price efficiency through the variance ratio test statistic captures only one dimension of efficiency (return predictability) and does not directly assess the speed or accuracy of price responses to specific information events. Fourth, the 29-observation sample, while adequate for ARDL estimation, limits the statistical power of the analysis and the ability to

conduct extensive robustness checks involving additional lag structures or structural break specifications. Fifth, the bid-ask spread measure used in this study is an annual average that may not fully capture the intraday and cross-sectional variation in spreads that microstructure theory emphasizes.

5.5 Future Research

Future research should exploit the increasing availability of higher-frequency data from the Nigerian Exchange Group to conduct daily or intraday analyses of the microstructure–efficiency nexus. The application of Markov-switching models or threshold regression techniques could capture potential regime-dependent relationships between microstructure and efficiency that the linear ARDL framework may not fully accommodate. Cross-country panel analyses comparing Nigeria with other African and emerging markets would provide valuable comparative evidence on the generalizability of the findings. Finally, the incorporation of measures of algorithmic and electronic trading activity, which have become increasingly relevant in the Nigerian market following the adoption of new trading technologies, would update the microstructural characterization and its efficiency implications.

6. Conclusion

6.1 Key Message

This study has demonstrated, through rigorous time series analysis, that market microstructure variables are powerful predictors of price efficiency in the Nigerian stock market. The evidence establishes that the informational efficiency of prices is not an innate property of markets but is shaped by the institutional, transactional, and structural environment within which trading occurs.

6.2 Key Research Findings

Three principal findings emerge from the analysis. First, the bid-ask spread is the single most important microstructural predictor of price efficiency, with wider spreads significantly reducing the degree to which prices approximate a random walk. This finding holds in both the short run and the long run, confirming the persistent nature of transaction cost barriers to efficiency. Second, trading volume and market depth positively predict price efficiency, supporting the theoretical proposition that greater market activity and absorptive capacity facilitate the incorporation of information into prices. Third, the macroeconomic and institutional environment matters profoundly: institutional quality, infrastructure development, trade openness, and economic growth enhance price efficiency, while natural resource dependence, inflation, and—to a lesser extent—weak human capital indicators detract from it.

6.3 Broader Implications

The broader implication of these findings is that market efficiency in emerging and frontier economies cannot be achieved through capital market reforms alone. It requires a holistic developmental approach that simultaneously addresses microstructural frictions, macroeconomic stability, institutional quality, and physical and human infrastructure. The interconnectedness of these factors means that partial reforms—addressing one

dimension while neglecting others—are unlikely to produce sustained improvements in price efficiency.

6.4 Main Research Contribution

The principal contribution of this study is the integration of market microstructure theory with the macroeconomic and institutional determinants of financial market outcomes within a unified, dynamic time series framework. By demonstrating that microstructural variables retain their predictive power after controlling for a comprehensive set of macroeconomic and institutional factors, the study establishes the independent, incremental explanatory role of market microstructure in determining price efficiency in a frontier market context. The methodological contribution lies in the application of the ARDL bounds testing approach to this research question in the Nigerian context, generating both short-run and long-run estimates with appropriate diagnostic validation.

6.5 Future Directions

Future scholarship should extend this analysis along several dimensions. The development of composite microstructure indices that capture multiple dimensions of market quality in a single metric would facilitate more parsimonious modeling. The examination of asymmetric effects—whether microstructural deterioration reduces efficiency more sharply than microstructural improvement enhances it—would provide nuanced policy insights. The investigation of spillover effects between the Nigerian stock market and other asset markets (bonds, foreign exchange, commodities) would illuminate the broader financial system implications of microstructural quality. The application of machine learning techniques to predict efficiency regimes based on microstructural and macroeconomic features represents a promising frontier that could enhance the practical relevance of academic research for market participants and regulators.

6.6 Call to Action

The findings of this study issue a clear call to action for Nigerian capital market regulators, policymakers, and market participants. The Securities and Exchange Commission of Nigeria and the Nigerian Exchange Group should prioritize the reduction of bid-ask spreads through enhanced market-making programs, the promotion of electronic trading technologies, and the reduction of regulatory barriers that constrain liquidity provision. Federal and state governments should recognize that capital market efficiency is not merely a financial sector concern but a development imperative that depends on macroeconomic stability, governance quality, infrastructure provision, and the diversification of the economy away from oil dependence. Market participants—institutional and retail investors, brokers, and asset managers—should engage constructively in the reform process, recognizing that improvements in market efficiency serve the collective interest by enhancing price discovery, reducing the cost of capital, and improving the allocation of resources across the Nigerian economy.

7. Recommendations

Based on the empirical findings, the following actionable recommendations are proposed:

- i. **Strengthen Market-Making Arrangements:** The Nigerian Exchange Group should expand and deepen its market-making program by providing incentives (reduced fees, favorable margin requirements) for designated market makers, with explicit obligations to maintain maximum bid-ask spreads for liquid securities. International evidence from the London Stock Exchange's reforms in the early 2000s demonstrates that well-designed market-making programs can reduce spreads by 30–40% within two years of implementation (Mayall, 2006).
- ii. **Enhance Trading Infrastructure and Technology:** Investment in advanced trading infrastructure, including improved matching engines, direct market access capabilities, and co-location services, can reduce latency and transaction costs. The adoption of continuous auction mechanisms for less liquid securities, supplemented by periodic call auctions, can improve price discovery for smaller-capitalization stocks.
- iii. **Promote Market Participation and Financial Literacy:** Broadening the investor base through targeted financial literacy campaigns, simplified account opening procedures, and the development of low-cost index investment products can increase trading volume and reduce the dominance of noise trading. The successful experience of India's National Stock Exchange in expanding retail participation through technology and education provides a relevant model (Shah & Thomas, 1998).
- iv. **Improve Institutional Quality and Regulatory Enforcement:** The fight against corruption and the strengthening of regulatory institutions are essential complements to capital market reform. The SEC Nigeria should be granted greater independence, adequate funding, and enhanced enforcement powers. The adoption of international best practices in corporate governance, disclosure requirements, and insider trading enforcement would signal institutional commitment to market integrity.
- v. **Pursue Economic Diversification:** Reducing the economy's dependence on oil revenue is a structural imperative with direct implications for capital market efficiency. Diversification into manufacturing, agriculture, technology, and services would expand the universe of listed companies, enhance the informational diversity of the market, and reduce the macroeconomic volatility associated with commodity price shocks.
- vi. **Invest in Physical and Digital Infrastructure:** Expanding access to electricity and internet connectivity—particularly in underserved regions—is essential for broadening market participation and improving the speed of information dissemination. The positive and significant effect of infrastructure development on price efficiency documented in this study underscores the complementarity between physical infrastructure investment and financial market development.
- vii. **Develop Human Capital:** While the direct effect of life expectancy on price efficiency was not statistically significant in this study, the theoretical linkage between human capital quality and financial market outcomes is well established. Investments in education, health, and technical skills development enhance the

- capacity of market participants to access, interpret, and act upon financial information, contributing to improved price discovery over the long term.
- viii. **Maintain Macroeconomic Stability:** The negative effect of inflation on price efficiency highlights the importance of sound monetary policy. The Central Bank of Nigeria should maintain its commitment to price stability, recognizing that macroeconomic uncertainty introduces noise into financial price signals and undermines the conditions necessary for efficient capital allocation.

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